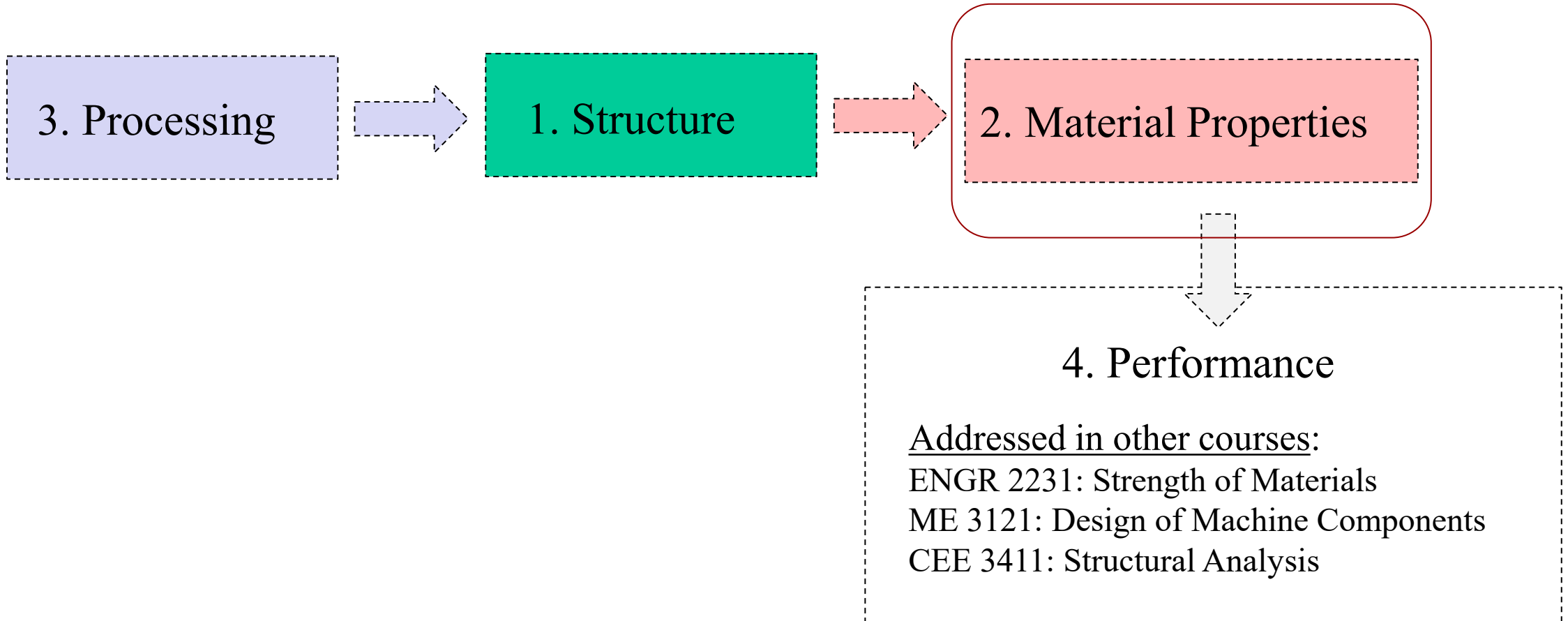


# Upcoming Dates

- **Sunday Night: Module 3 – Video**
- **Tuesday Night: Module 3 – Watch Video + Post lab worksheet**
- **Exam 1 – Thursday March 5**

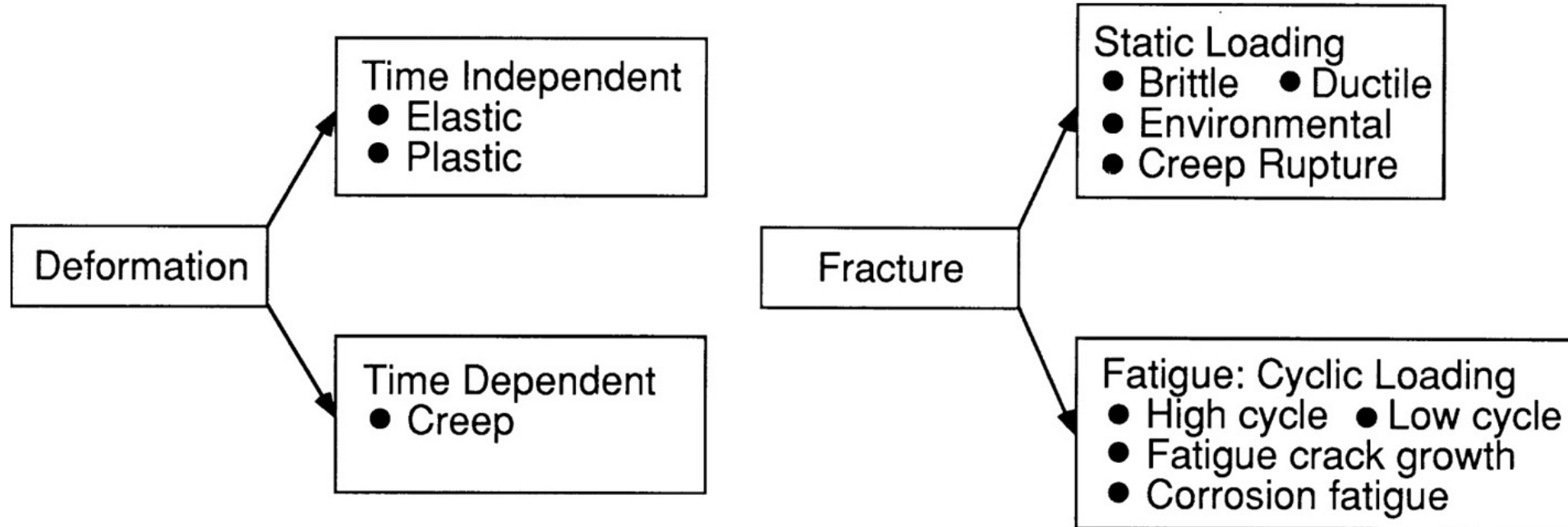
# Today's Lecture

We are here!

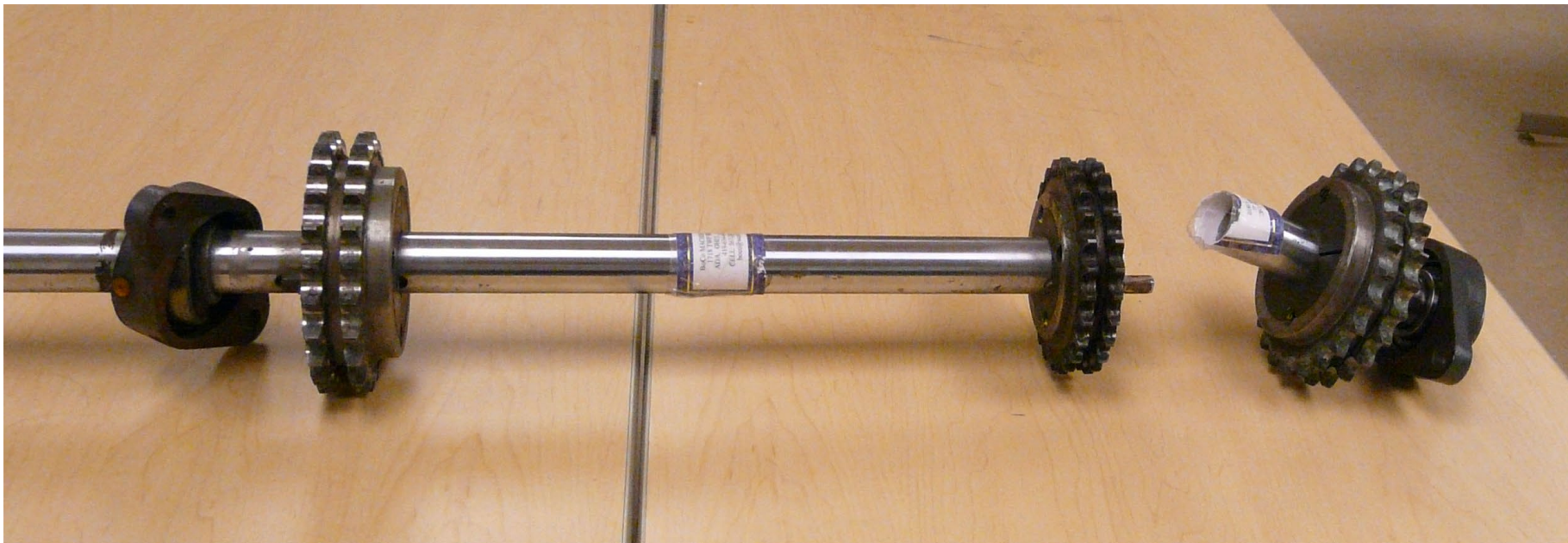


# CH 6 & 7 Properties of Solid Materials

## Mechanical Behavior of Materials

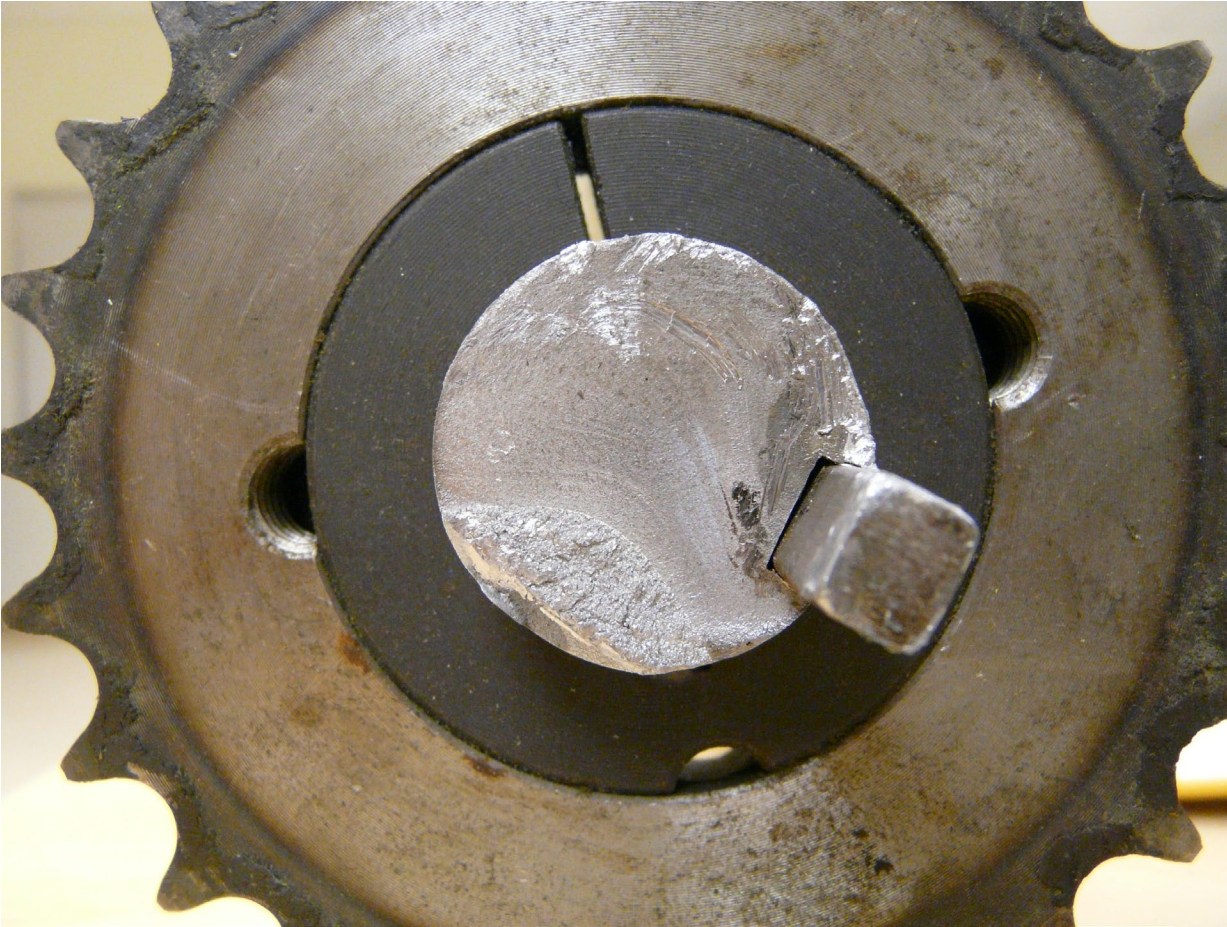


# Broken shaft from Proctor & Gamble





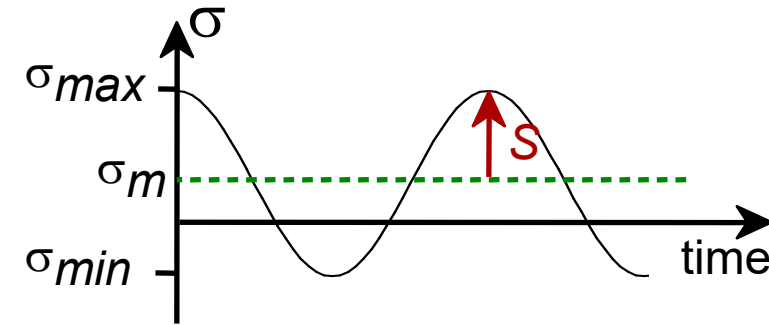
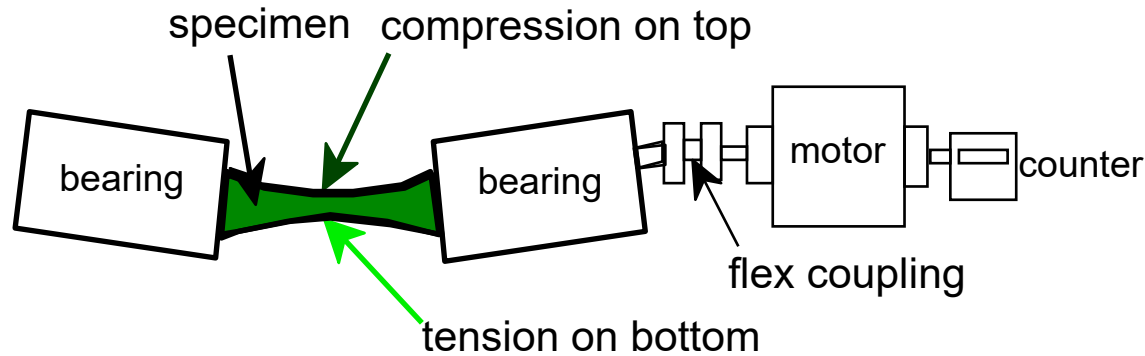
# The fracture surface



# Fatigue

- The “beachmark ridges” are indicators of fatigue.
- Fatigue occurs in structures subjected to dynamic and fluctuating stresses, it is possible for failure to occur at a stress level considerably lower than the tensile or yield strength for a static load.
- As the shaft experiences an unbalanced cyclical load, over time the failure crack initiates from a weak spot where there is stress concentration

# Cyclical Stresses Result in Fatigue Failure

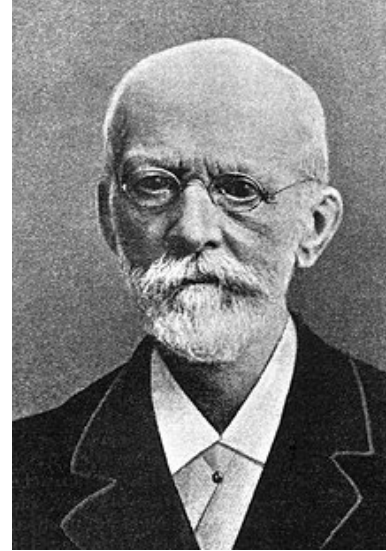
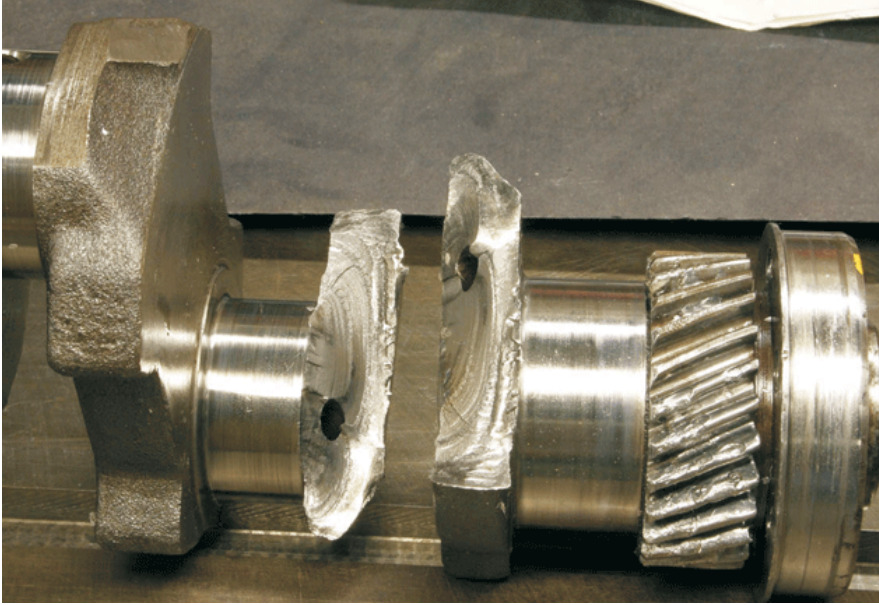


- Stress varies with time.
  - $S \rightarrow$  Stress Amplitude or Alternating Stress
  - $\sigma_m \rightarrow$  Mean Stress
  - Frequency  $\rightarrow$  cycles/time
- Fatigue Failure occurs due to cyclical stresses
  - can cause part failure, even though  $\sigma_{max} < \sigma_{uts}$ .
  - causes ~ 90% of mechanical engineering failures.

Adapted from Fig. 9.24,  
Callister & Rethwisch 3e.  
(Fig. 9.24 is from *Materials  
Science in Engineering*, 4/E  
by Carl. A. Keyser, Pearson  
Education, Inc., Upper  
Saddle River, N.J.)



# Fatigue - Historically



August Wöhler

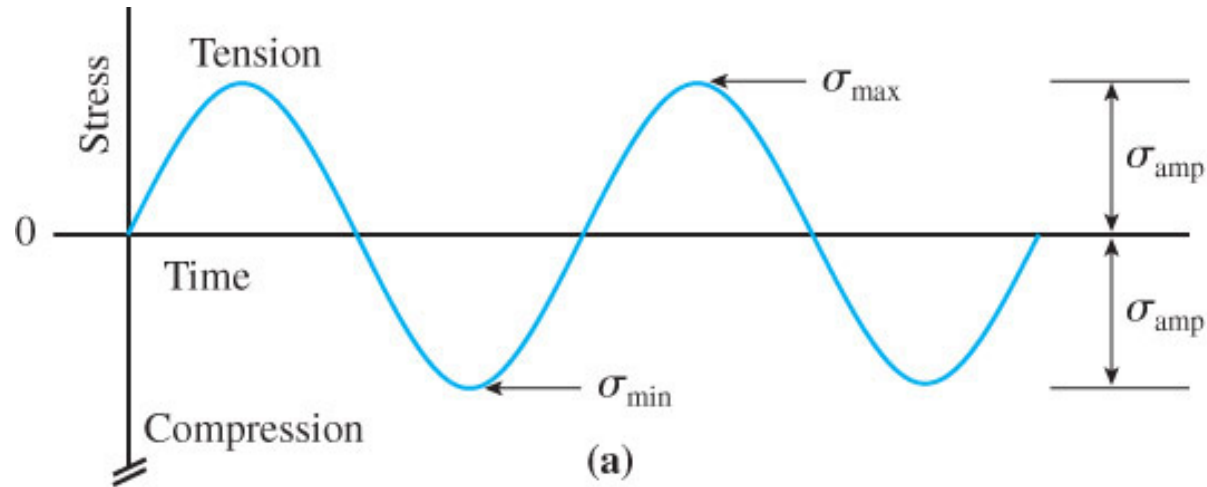


Jean-Victor Poncelet

- The phenomenon of fatigue failure was first noticed in the 1800s when railroad-car axles began failing at stresses much below the tensile strength.
- Sudden, brittle failures
- Called “Fatigue” because it was originally hypothesized that the material had somehow become “tired” and weaker.



# Constant Amplitude Cycles



**stress amplitude:**

$$\sigma_a = S = \frac{\sigma_r}{2} = \frac{\sigma_{max} - \sigma_{min}}{2}$$

**mean stress:**  $\sigma_m = \frac{\sigma_{max} + \sigma_{min}}{2}$

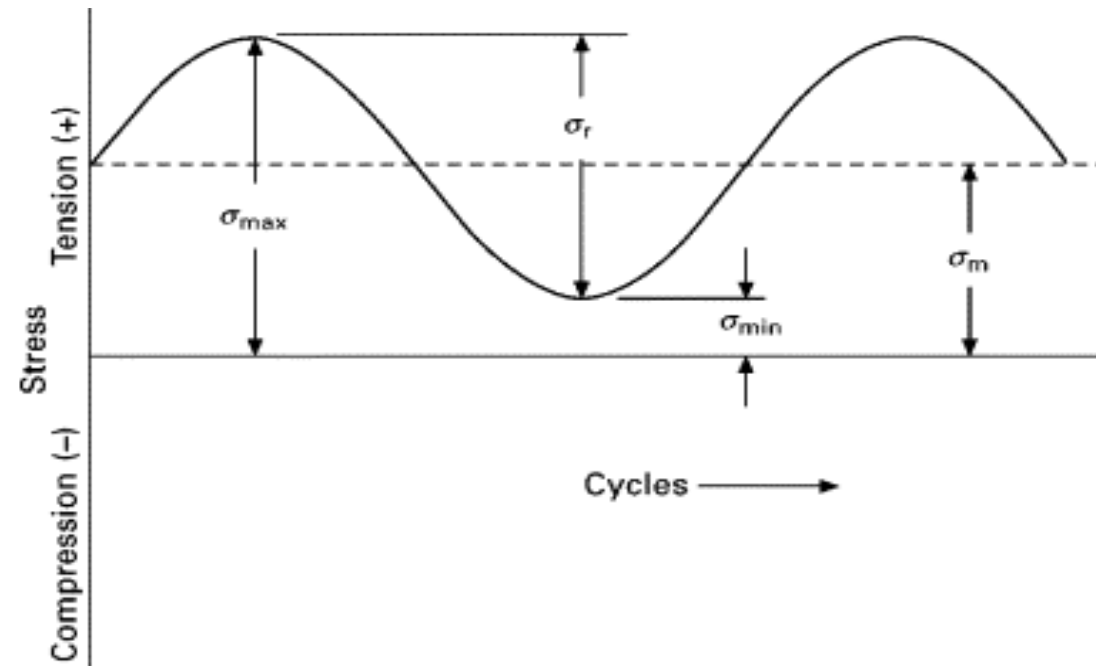
**stress range:**  $\sigma_r = \sigma_{max} - \sigma_{min}$

**Note:** the stress we compare to the endurance limit is combination of stress amplitude and mean stress. (ME 3121 – Design of Machine Components)

# Example

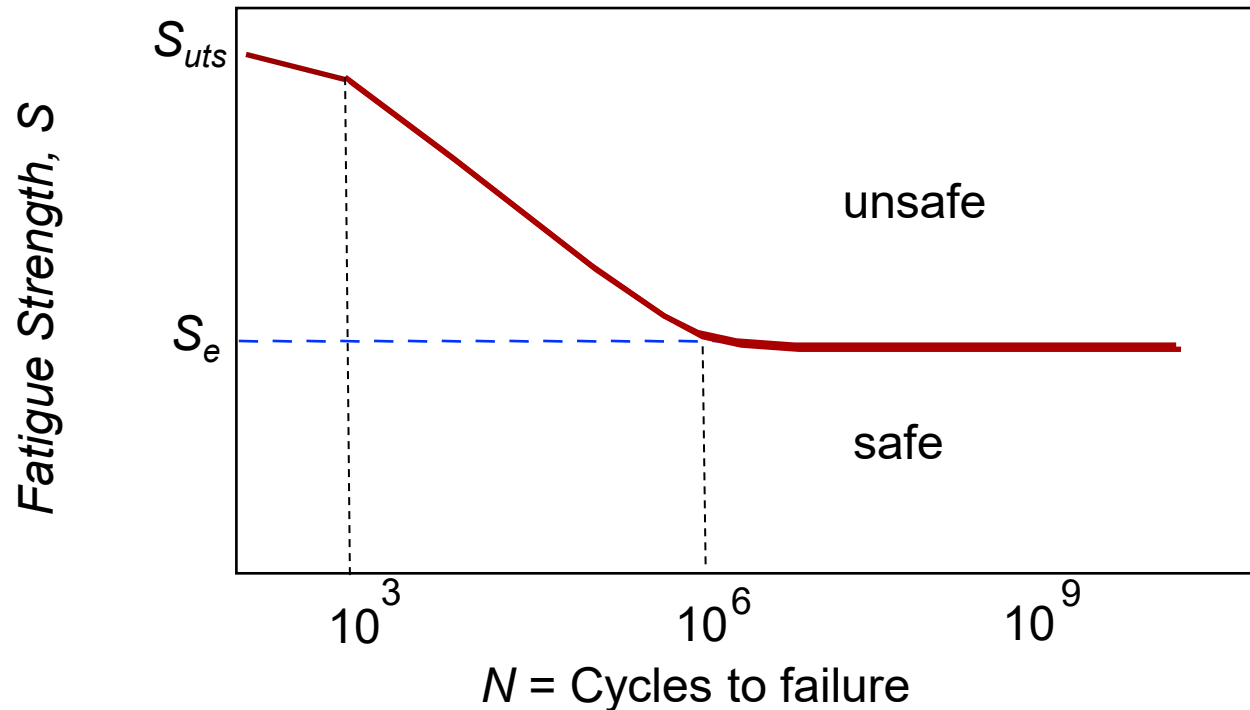
For the following simple constant amplitude cyclic stress is shown below, the magnitude of the stress varies between a constant maximum  $\sigma_{\max}$  and a constant minimum  $\sigma_{\min}$  with a value of 500 MPa and 100 MPa.

Determine The stress range, mean stress, and stress amplitude are



# S-N curve:

## Material's susceptibility to fatigue

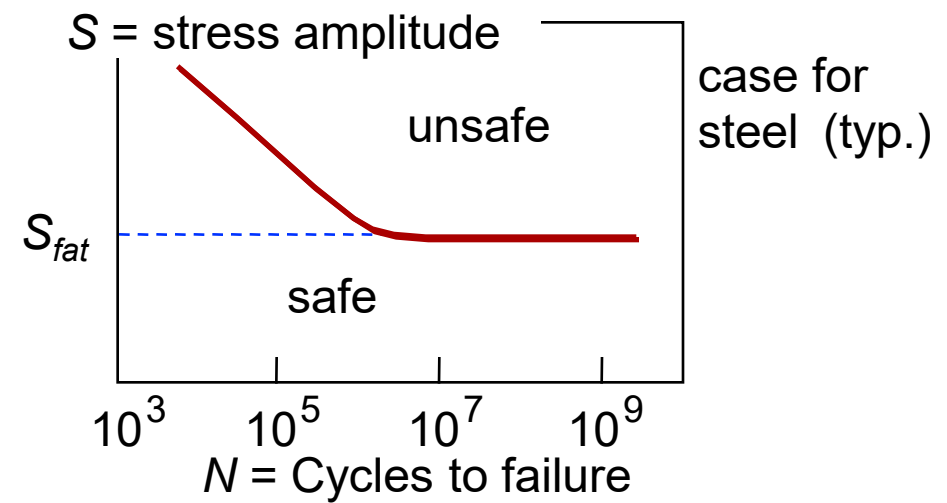


Adapted from Fig. 9.25(a), *Callister & Rethwisch* 3e.

- Fatigue limit ( $S_{fat}$ ) or Endurance Limit ( $S_e$ )
  - No fatigue failure if  $\sigma < S_e$
- For steels, some other ferrous (iron based) and titanium alloys:
  - Infinite Life  $\rightarrow 10^6$  cycles

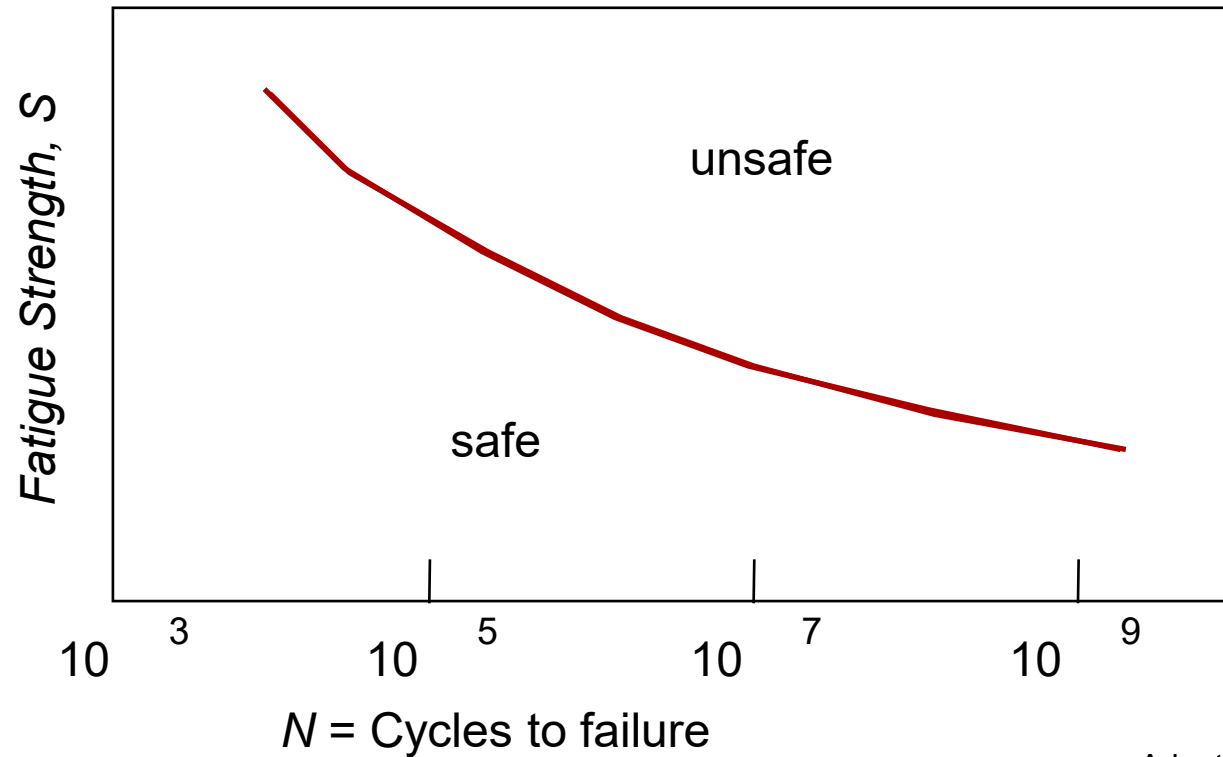


It's a bumpy road people but  
just keep going!!





# Most Non Ferrous Alloys Don't have an Endurance Limit!

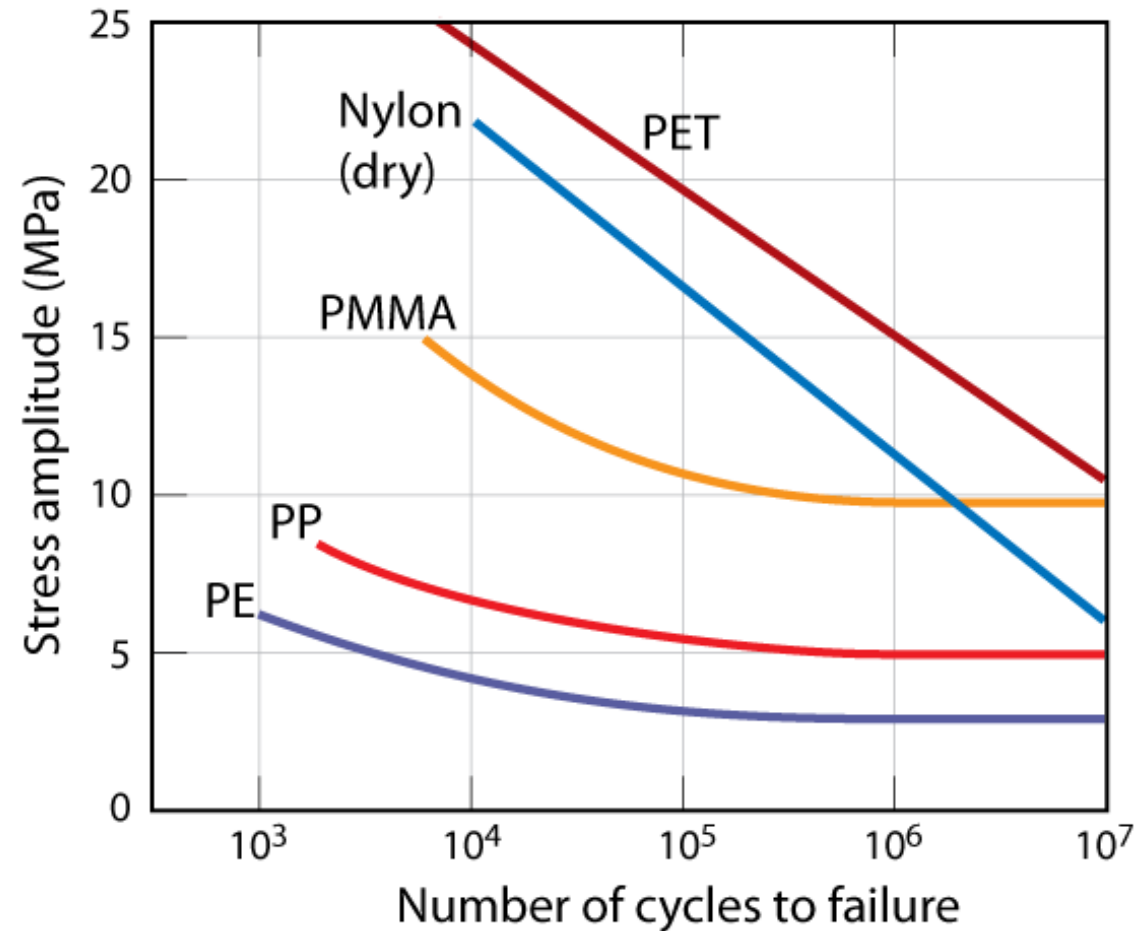


Adapted from Fig. 9.25(b), *Callister & Rethwisch 3e*.

- For most nonferrous alloys such as Al, Cu, Mg
- Design goal:  $5 \times 10^8$  cycles

# Fatigue Behavior of Polymers

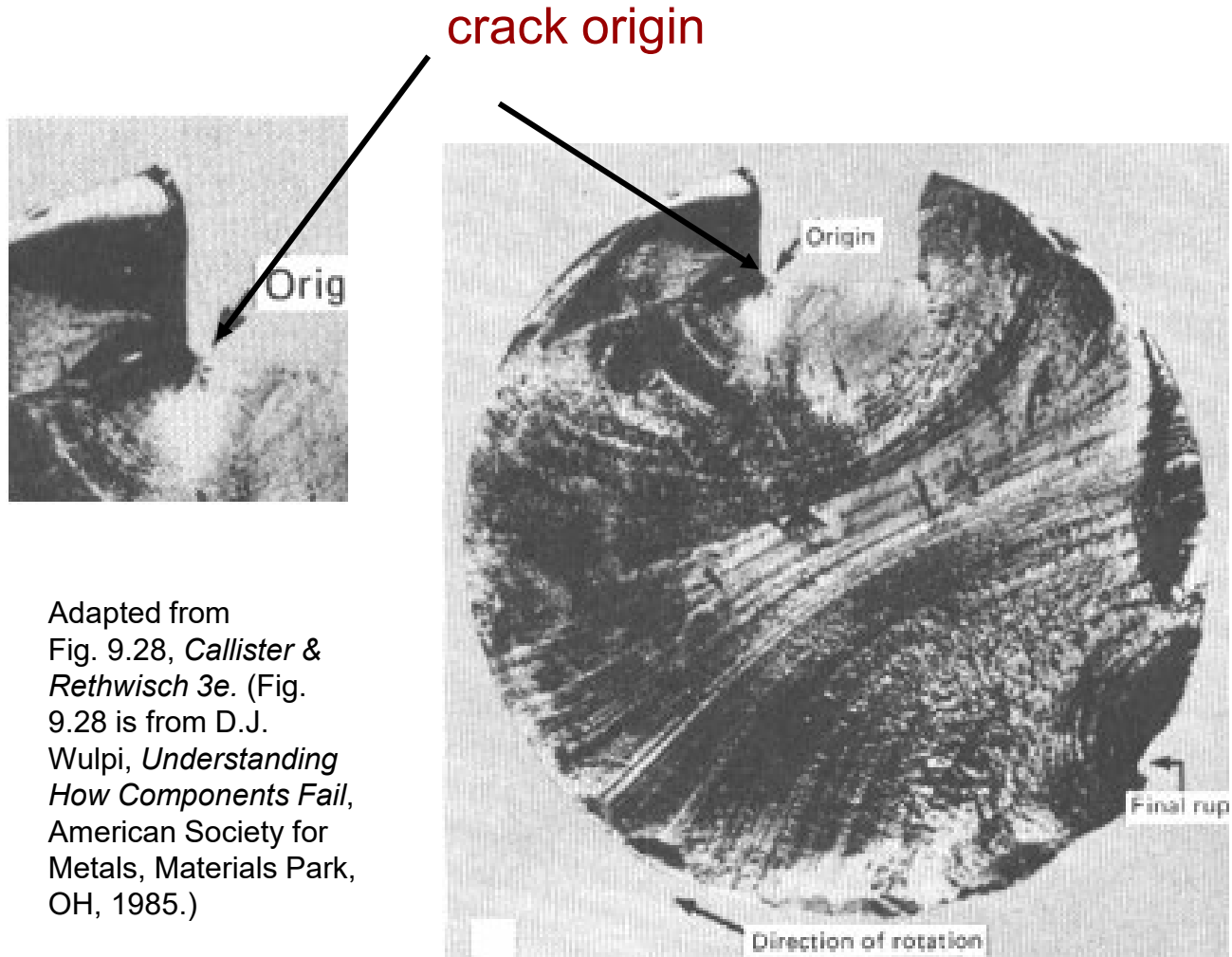
- Fatigue limit:
  - PMMA, PP, PE
- No fatigue limit:
  - PET, Nylon (dry)



Adapted from Fig. 9.27,  
*Callister & Rethwisch 3e.*

# Fatigue Mechanism

- Crack grows *incrementally*
- Failed rotating shaft
  - crack grew even though  $K_{max} < K_c$
  - crack grows faster as
    - $\Delta\sigma$  increases
    - crack gets longer
    - loading freq. increases.
    - tensile mean stress

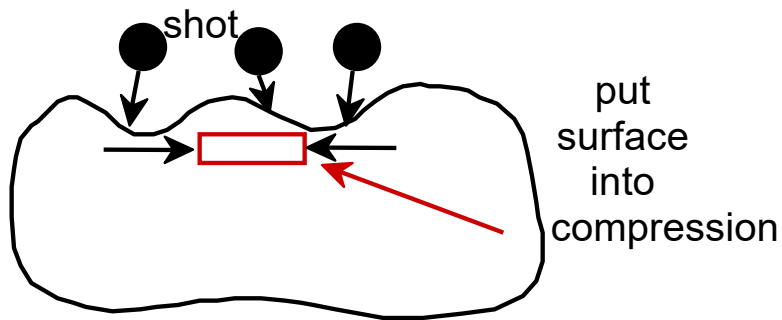


Adapted from  
Fig. 9.28, *Callister &  
Rethwisch 3e*. (Fig.  
9.28 is from D.J.  
Wulpi, *Understanding  
How Components Fail*,  
American Society for  
Metals, Materials Park,  
OH, 1985.)

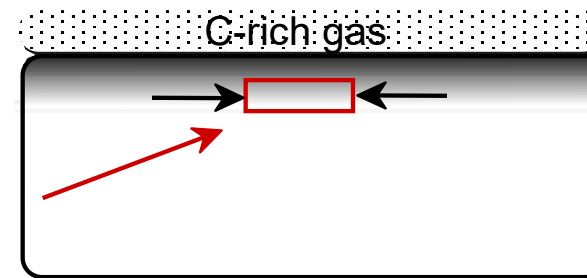
# Improving Fatigue Life

1. Impose a compressive surface stress(to suppress surface cracks from growing)

--Method 1: shot peening

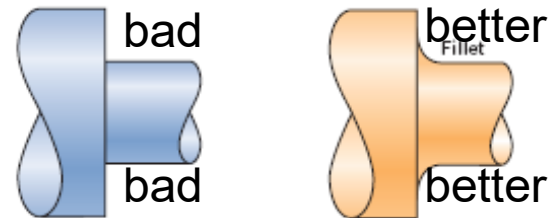


--Method 2: carburizing



Adapted from Fig. 9.31, *Callister & Rethwisch 3e*.

2. Remove stress concentrators.



Adapted from Fig. 9.32, *Callister & Rethwisch 3e*.

# Improving Fatigue Life

**3. Polish surface**

**4. Avoid tensile residual stress (eg. Composite)**



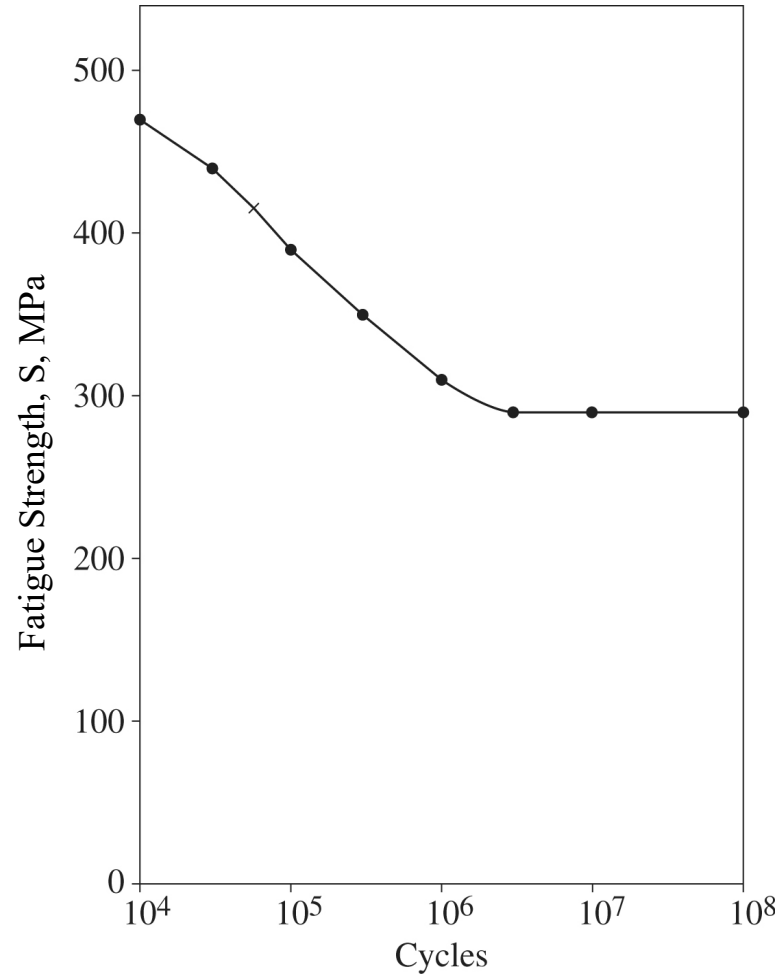
# Examples

Stress amplitude  
MPa(ksi)

470(68.0)  
440(63.4)  
390(56.2)  
350(51.0)  
310(45.3)  
290(42.2)  
290(42.2)  
290(42.2)

Cycles to failure

$10^4$   
 $3 \times 10^4$   
 $10^5$   
 $3 \times 10^5$   
 $10^6$   
 $3 \times 10^6$   
 $10^7$   
 $10^8$



1. Life at  $\sigma = 440$  MPa

2. S at  $10^5$  cycles

3. Life at  $\sigma = 100$  Mpa

4. Endurance Limit,  $S_e$

# Chapter 7.10: Time Dependent Strain Phenomena

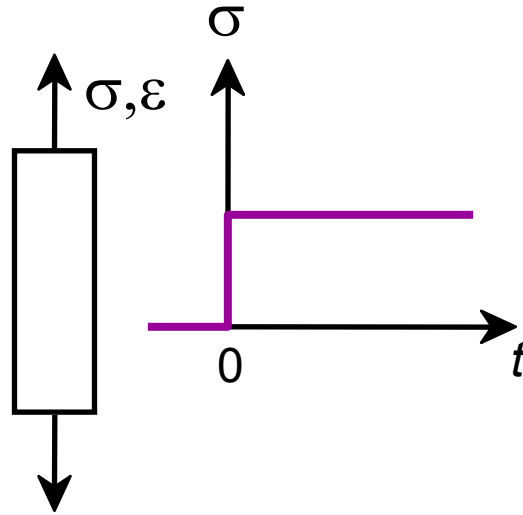


©EVOX IMAGES

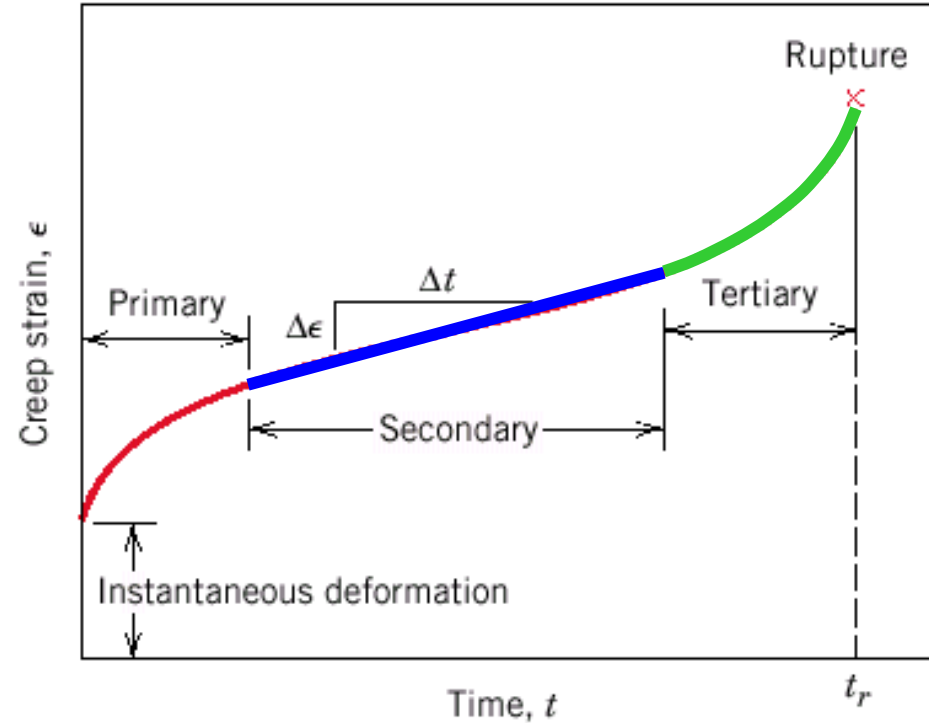
**After a car door has been left open for a long time, the door can not be closed properly. However, no corrosion is observed. What causes this problem?**

**The hinges creep!**

# Sample deformation at a constant stress ( $\sigma$ ) vs. time



## Creep Occurs!



Adapted from Fig. 9.35, Callister & Rethwisch 3e.

- Primary Creep: slope (creep rate) decreases with time.
- Secondary Creep: steady-state
  - i.e., constant slope.
- Tertiary Creep: slope (creep rate) increases with time,
  - i.e. acceleration of rate.

# Creep Examples



# Possible Mechanisms

## Metals

- 1) diffusional flow of vacancies
- 2) sliding of grain boundary

## Polymers

- 1) molecular chains in polymer slide past one another in a time-dependent manner
- 2) viscous flow

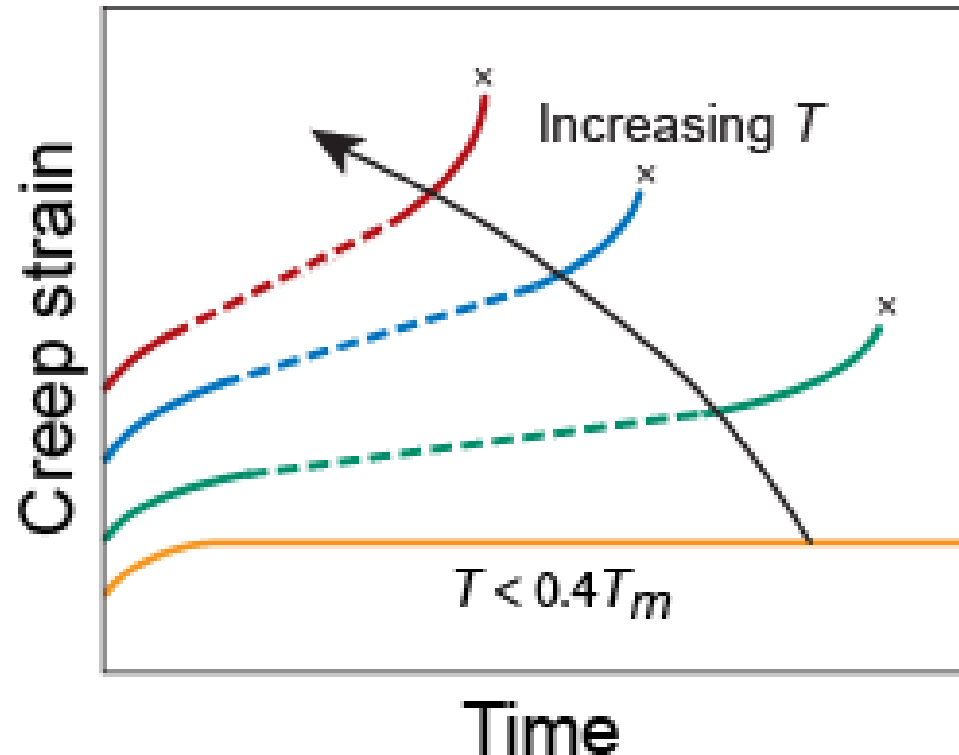
# Time-dependent Strain Phenomenon:

*As temperature increases, creep strain increases (creep faster)*

*According to The Essence of Material for Engineers (textbook):*

*Metals:  $T > 0.4 T_m$  (melting temperature in Kelvin)*

*Polymers:  $T > T_g$  (glass transition temperature)*



Adapted from Figs. 9.36,  
Callister & Rethwisch 3e.



## From other sources:

- As a general rule, creep occurs when metals are required to operate at temperatures above 30 to 40% of their absolute melting point (in Kelvin).
- Rapid creep of polymers occurs at 30–40% of their glass transition temperature (in Kelvin).
- Creep of heat-resistant ceramic materials starts above 40–50% of their melting temperature (in Kelvin).

***Kelvin to Celsius?  $C = K - 273.15$***

# Public Toilet Partition Door Lock



# Which design considers the impact of time?



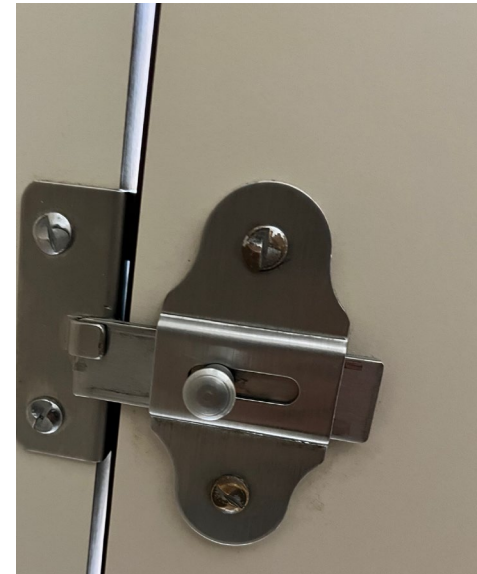
(Design A)



(Design B)



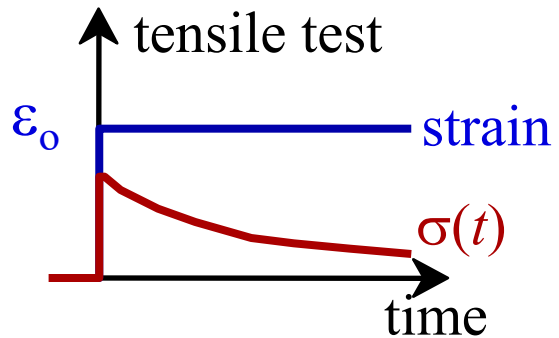
(Design C)



## 2. Stress Relaxation in Polymers

### Stress relaxation test:

- strain in tension to  $\epsilon_0$  and hold.
- observe decrease in stress with time.

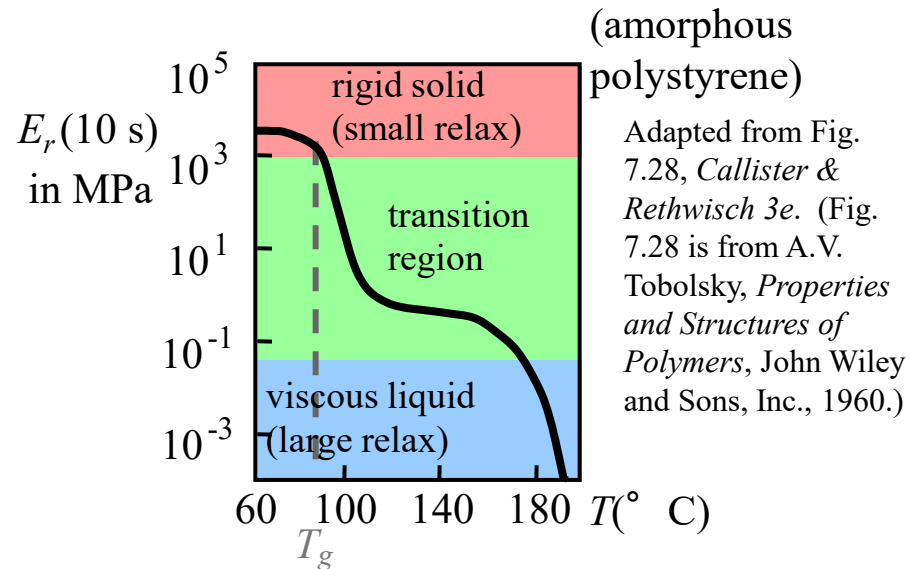


Relaxation modulus:

$$E_r(t) = \frac{\sigma(t)}{\epsilon_0}$$

# Time-Dependent Deformation

There is a large decrease in  $E_r$  for  $T > T_g$ .



## Representative $T_g$ values (°C):

|                   |       |
|-------------------|-------|
| PE (low density)  | - 110 |
| PE (high density) | - 90  |
| PVC               | + 87  |
| PS                | +100  |
| PC                | +150  |

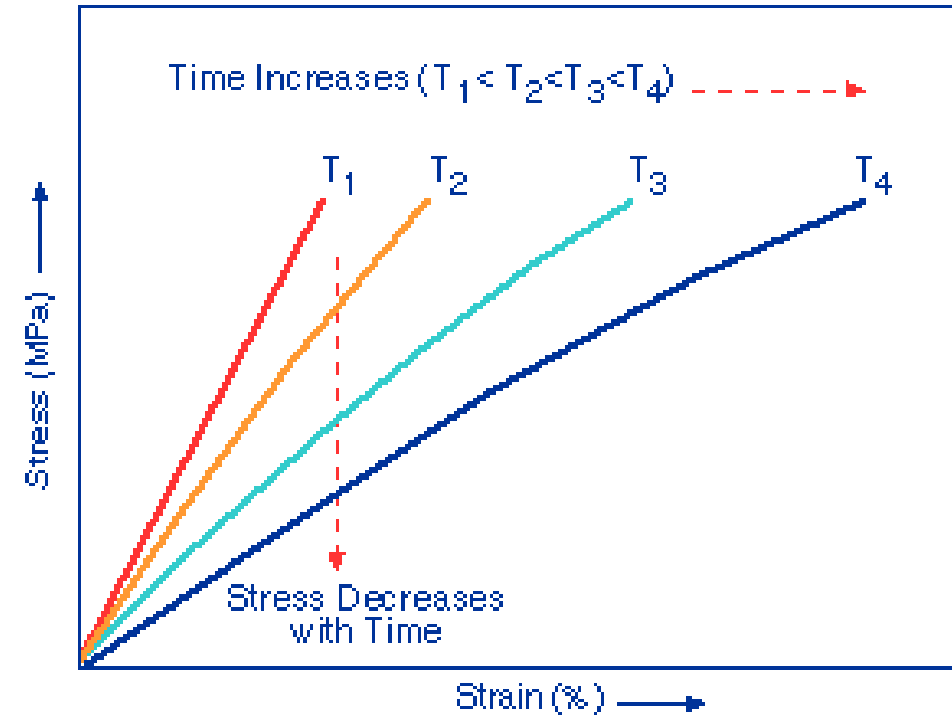
Selected values from  
Table 11.3, Callister &  
Rethwisch 3e.



**Silly putty --- bounce when dropped, and could stretch farther than regular rubber**

# Influence of $T$ and Strain Rate

- Decreasing  $T$ ...
  - increases  $E$
  - increases  $S_{uts}$
  - decreases  $\%EL$
- Increasing strain rate...
  - same effects as decreasing  $T$ .



## Examples - Temperature

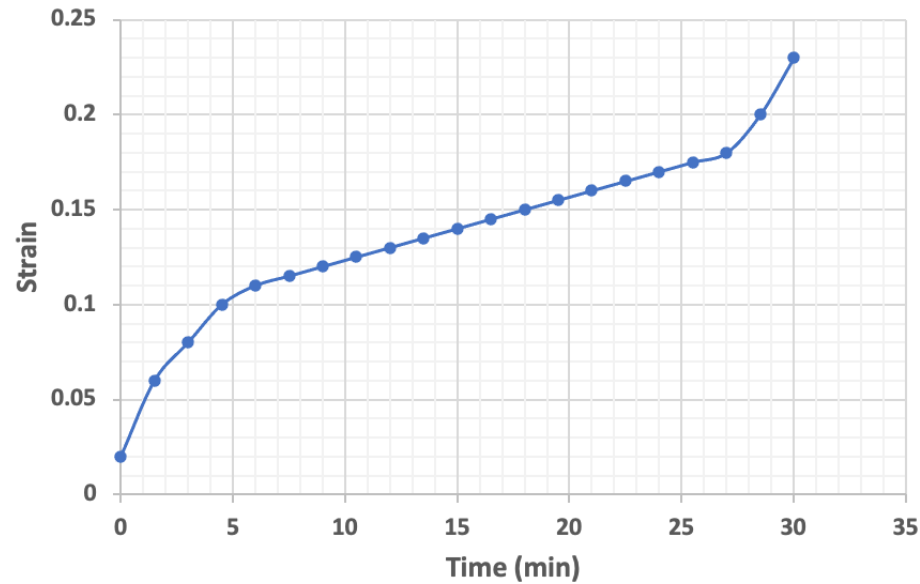
**The melting temperature of pure aluminum is 660.3°C, above about what temperature does creep become an important failure mechanism for aluminum?**

**The melting temperature of pure iron is 1538°C, above about what temperature does creep become an important failure mechanism for iron?**



# Example

| Time (min) | Strain |
|------------|--------|
| 0          | 0.02   |
| 1.5        | 0.06   |
| 3          | 0.08   |
| 4.5        | 0.1    |
| 6          | 0.11   |
| 7.5        | 0.115  |
| 9          | 0.12   |
| 10.5       | 0.125  |
| 12         | 0.13   |
| 13.5       | 0.135  |
| 15         | 0.14   |
| 16.5       | 0.145  |
| 18         | 0.15   |
| 19.5       | 0.155  |
| 21         | 0.16   |
| 22.5       | 0.165  |
| 24         | 0.17   |
| 25.5       | 0.175  |
| 27         | 0.18   |
| 28.5       | 0.2    |
| 30         | 0.23   |



1. What is the steady-state creep rate of the metal?
2. What is the creep rupture time ( $t_r$ ) of the metal?